



IT-SIMPLERA INSTITUTE -CYBER SECURITY ANALYST INTERNSHIP PROGRAM

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Executive Summary

Computer networking serves as the backbone of modern information technology and cybersecurity. Every digital communication, from browsing websites to accessing cloud services, relies on secure and efficient network infrastructure. Understanding how computers communicate across local and wide-area networks is essential for cybersecurity professionals responsible for protecting organizational assets.

This engagement report presents practical and theoretical activities completed during Week 02 of the IT-SIMPLERA Cyber Security Analyst Internship Program. The exercises focused on identifying computer network configuration, executing essential network diagnostic commands, comparing networking devices, analyzing Internet Protocol versions, and understanding the Open Systems Interconnection (OSI) Reference Model.

Practical activities included using the **ipconfig /all**, **ping**, **tracert**, **nslookup**, and **arp** commands to examine network configurations, verify connectivity, troubleshoot communication paths, resolve domain names, and inspect Address Resolution Protocol entries. These tools are widely used by network administrators, system engineers, and cybersecurity analysts for troubleshooting and incident response.

The report also examines the functional differences between routers and switches, compares IPv4 and IPv6 addressing technologies, and explains each layer of the OSI Reference Model together with common protocols and networking devices.

Overall, the knowledge and practical experience gained through this assignment provide a strong foundation for future studies in network administration, digital forensics, penetration testing, and security operations.

2. Learning Objectives

The primary objective of this internship assignment is to develop a strong understanding of fundamental computer networking concepts and practical network configuration techniques that are essential for cybersecurity professionals.

Upon successful completion of this assignment, the following learning objectives were achieved:

1. Understand the basic principles of computer networking and data communication.
2. Identify and interpret important network configuration information, including IPv4 Address, MAC Address, Default Gateway, DNS Server, and Subnet Mask.
3. Gain practical experience using Windows networking commands to troubleshoot and analyze network connectivity.
4. Explain the purpose and functionality of common networking utilities including **ipconfig**, **ping**, **tracert**, **nslookup**, and **arp**.
5. Differentiate between routers and switches based on their functions, operating layers, addressing methods, and practical applications.
6. Compare IPv4 and IPv6 addressing schemes, highlighting their architecture, advantages, and limitations.
7. Describe each layer of the OSI Reference Model, including its responsibilities, associated protocols, and networking devices.
8. Improve technical documentation skills by presenting networking information in a clear, organized, and professional manner.

3.1 Task 01 – Network Information

Objective

The purpose of this task is to examine the computer's network configuration using the **ipconfig /all** command. This command provides detailed information about all network adapters installed on the system and displays configuration details that are essential for troubleshooting and network administration.

Command Executed

Ipconfig /all

```
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>shutdown /h

C:\Users\user>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : aliyubacceitas
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Mixed
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : BC-0F-F3-D4-39-9C
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : ExpressVPN TAP Adapter
    Physical Address. . . . . : 00-FF-A1-D2-CE-F1
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Unknown adapter Local Area Connection:
```

Network Configuration Analysis

Network Parameter	Description	Purpose
IPv4 Address	Unique logical address assigned to the computer within the local network.	Enables communication with other devices on the same network and across the Internet.
Physical (MAC) Address	Permanent hardware identifier assigned by the network interface manufacturer.	Allows devices on the local network to uniquely identify one another at the Data Link Layer.
Default Gateway	IP address of the router that connects the local network to external networks.	Directs outgoing traffic toward other networks, including the Internet.

Additional Network Information

Field	Explanation
Host Name	Identifies the computer on the network.
DNS Suffix	Specifies the domain associated with the network.
DHCP Enabled	Indicates whether the IP address is assigned automatically by a DHCP server.
Subnet Mask	Defines the network and host portions of the IP address.
DNS Server	Resolves domain names into IP addresses.
Lease Obtained	Indicates the date and time when the IP address was assigned.

Field	Explanation
Lease Expires	Shows when the DHCP lease will expire.

Discussion

The **ipconfig /all** command is one of the most frequently used diagnostic utilities in Microsoft Windows. It enables network administrators and cybersecurity professionals to verify IP addressing, troubleshoot connectivity issues, identify DHCP configuration, and confirm DNS server settings. Understanding this information is fundamental when diagnosing network problems or investigating security incidents.

3.2 Task 02 – Basic Networking Commands

Network troubleshooting requires several built-in command-line utilities that help administrators verify connectivity, identify routing issues, resolve domain names, and inspect network communications. The following commands were executed during this exercise.

1. ipconfig /all

Command

```
ipconfig /all
```

Purpose

Displays detailed TCP/IP configuration information for all network interfaces installed on the computer.

Figure 3.2.1 – ipconfig /all Output

```

Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>shutdown /h

C:\Users\user>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : aliyubacceitas
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Mixed
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : BC-0F-F3-D4-39-9C
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : ExpressVPN TAP Adapter
    Physical Address. . . . . : 00-FF-A1-D2-CE-F1
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Unknown adapter Local Area Connection:

```

2. ping google.com

Command

ping google.com

Purpose

The **ping** command sends Internet Control Message Protocol (ICMP) Echo Request packets to a remote host and measures the time required for the responses. It is commonly used to verify network connectivity and measure lat

Figure 3.2.2 – Ping Command Output

```
C:\Users\user>ping google.com

Pinging google.com [172.217.16.238] with 32 bytes of data:
Reply from 172.217.16.238: bytes=32 time=180ms TTL=109
Reply from 172.217.16.238: bytes=32 time=170ms TTL=109
Reply from 172.217.16.238: bytes=32 time=162ms TTL=109
Reply from 172.217.16.238: bytes=32 time=152ms TTL=109

Ping statistics for 172.217.16.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 152ms, Maximum = 180ms, Average = 166ms

C:\Users\user>|
```

3. tracert google.com

Command

tracert google.com

Purpose

The **tracert** command identifies the sequence of routers (known as hops) through which network packets travel to reach a destination. This command is useful for locating delays or communication failures along the transmission path

Figure 3.2.3 – Tracert Command Output

```
C:\Users\user>tracert

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
              [-R] [-S srcaddr] [-4] [-6] target_name

Options:
    -d                Do not resolve addresses to hostnames.
    -h maximum_hops   Maximum number of hops to search for target.
    -j host-list       Loose source route along host-list (IPv4-only).
    -w timeout         Wait timeout milliseconds for each reply.
    -R                Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                Force using IPv4.
    -6                Force using IPv6.

C:\Users\user>|
```

4. nslookup google.com

Command

nslookup google.com

Purpose

The **nslookup** utility queries the Domain Name System (DNS) to resolve a domain name into its corresponding IP address or perform reverse lookups. It assists in diagnosing DNS-related issues.

Screenshot

Figure 3.2.4 – NSLookup Output

```
C:\Users\user>ping google.com

Pinging google.com [172.217.16.238] with 32 bytes of data:
Reply from 172.217.16.238: bytes=32 time=180ms TTL=109
Reply from 172.217.16.238: bytes=32 time=170ms TTL=109
Reply from 172.217.16.238: bytes=32 time=162ms TTL=109
Reply from 172.217.16.238: bytes=32 time=152ms TTL=109

Ping statistics for 172.217.16.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 152ms, Maximum = 180ms, Average = 166ms

C:\Users\user>
```

5. arp -a

Command

arp -a

Purpose

The **arp -a** command displays the Address Resolution Protocol (ARP) cache stored by the operating system. The ARP table maps IPv4 addresses to their corresponding MAC addresses, allowing devices on the local network to communicate efficiently.

Figure 3.2.5 – ARP Table Output

```
C:\Users\user>arp
```

Displays and modifies the IP-to-Physical address translation tables used by address resolution protocol (ARP).

```
ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]
```

-a Displays current ARP entries by interrogating the current protocol data. If `inet_addr` is specified, the IP and Physical addresses for only the specified computer are displayed. If more than one network interface uses ARP, entries for each ARP table are displayed.

-g Same as **-a**.

-v Displays current ARP entries in verbose mode. All invalid entries and entries on the loop-back interface will be shown.

inet_addr Specifies an internet address.

-N if_addr Displays the ARP entries for the network interface specified by `if_addr`.

-d Deletes the host specified by `inet_addr`. `inet_addr` may be wildcarded with `*` to delete all hosts.

-s Adds the host and associates the Internet address `inet_addr` with the Physical address `eth_addr`. The Physical address is given as 6 hexadecimal bytes separated by hyphens. The entry is permanent.

eth_addr Specifies a physical address.

if_addr If present, this specifies the Internet address of the interface whose address translation table should be modified.

Summary of Networking Commands

Command	Primary Function	Common Use Case
ipconfig /all	Displays complete network configuration.	Verify IP settings and troubleshoot network adapters.
ping	Tests connectivity using ICMP packets.	Confirm that a remote host is reachable.
tracert	Displays packet routing path.	Diagnose routing problems and network delays.
nslookup	Queries DNS servers.	Verify DNS resolution and domain records.

Command	Primary Function	Common Use Case
arp -a	Displays the local ARP cache.	View IP-to-MAC address mappings on the local network.

3.3 Task 03 – Router vs. Switch

Objective

The purpose of this task is to understand the differences between a router and a switch, two essential networking devices used in modern computer networks. Although both devices facilitate communication, they perform different functions and operate at different layers of the OSI Reference Model.

Router

A **router** is a Layer 3 (Network Layer) device responsible for connecting multiple networks together. It forwards data packets between different networks using IP addresses and selects the most efficient path for data transmission. Routers are commonly used to connect Local Area Networks (LANs) to the Internet and to enable communication between different subnets.

Common Uses of a Router

- Connects a home or office network to the Internet.
- Routes traffic between different networks.
- Performs Network Address Translation (NAT).
- Supports firewall and security features.
- Enables communication between branch offices through Wide Area Networks (WANs).

Switch

A **switch** is primarily a Layer 2 (Data Link Layer) device that connects multiple devices within the same Local Area Network (LAN). It forwards data frames based on MAC addresses, ensuring that data reaches the correct destination efficiently. Modern enterprise switches may also support Layer 3 functions such as inter-VLAN routing.

Common Uses of a Switch

- Connects computers within an office network.
- Connects printers, servers, and IP phones.
- Supports Virtual Local Area Networks (VLANs).
- Reduces unnecessary network traffic.
- Improves network performance by minimizing collisions.

Router vs. Switch Comparison

Comparison Point	Router	Switch
Primary Function	Connects different networks	Connects devices within the same network
OSI Layer	Layer 3 (Network Layer)	Layer 2 (Data Link Layer)
Address Used	IP Address	MAC Address
Data Unit	Packet	Frame
Broadcast Domain	Separates broadcast domains	Shares a broadcast domain unless VLANs are configured
Collision Domain	Each interface has its own collision domain	Each port has its own collision domain
Speed	Generally slower due to routing decisions	Generally faster because of hardware switching
Security Features	Supports NAT, ACLs, VPNs, and firewall functions	Supports VLANs, Port Security, and MAC filtering
Typical Deployment	Internet gateway, branch office connectivity	Office LANs, campus networks, data centers
Cost	Typically, more expensive	Typically, less expensive

Key Differences

- A router connects **multiple networks**, whereas a switch connects **multiple devices within a single network**.
- Routers make forwarding decisions using **IP addresses**, while switches use **MAC addresses**.
- Routers are responsible for directing Internet traffic, whereas switches provide efficient communication inside a LAN.
- Routers often include advanced security features such as NAT, firewall capabilities, and VPN support. Switches focus on efficient data forwarding and network segmentation through VLANs.

Conclusion

Routers and switches are complementary networking devices. A switch builds the internal network by connecting local devices, while a router enables communication between different networks and the Internet. Together, they form the foundation of modern enterprise and home network infrastructures.

3.4 Task 04 – IPv4 vs. IPv6

Objective

Internet Protocol (IP) provides logical addressing for devices communicating over a network. Due to the rapid growth of Internet-connected devices, IPv6 was introduced to overcome the limitations of IPv4 and to provide a significantly larger address space.

IPv4 Overview

Internet Protocol Version 4 (IPv4) is the most widely deployed version of the Internet Protocol. It uses a **32-bit address**, allowing approximately **4.3 billion unique addresses**. IPv4 addresses are represented in dotted-decimal notation.

Example IPv4 Address

192.168.1.25

IPv6 Overview

Internet Protocol Version 6 (IPv6) is the successor to IPv4. It uses a **128-bit address**, providing an extremely large address space capable of supporting billions of connected devices worldwide. IPv6 addresses are represented in hexadecimal notation separated by colons.

Example IPv6 Address

2001:0db8:85a3:0000:0000:8a2e:0370:7334

IPv4 vs. IPv6 Comparison

Feature	IPv4	IPv6
Address Length	32-bit	128-bit
Address Format	Decimal numbers separated by periods	Hexadecimal numbers separated by colons
Address Example	192.168.1.25	2001:db8:85a3::8a2e: 370:7334
Number of Addresses	Approximately 4.3 billion	Approximately 340 undecillions
Header Size	Variable	Fixed
Configuration	Manual or DHCP	SLAAC or DHCPv6
Broadcast Support	Yes	No (uses multicast and anycast)
Security	IPSec optional	IPSec support built into the standard
Address Conservation	Uses NAT extensively	NAT generally not required
Future Scalability	Limited	Designed for future Internet growth

Advantages of IPv6

- Vastly larger address space.
- Simplified packet header for efficient routing.
- Improved support for mobile devices.
- Better multicast communication.
- Automatic address configuration (SLAAC).
- Enhanced support for end-to-end connectivity.
- Improved scalability for IoT and cloud environments.

Conclusion

Although IPv4 remains the dominant protocol in many networks, IPv6 is increasingly being adopted due to its scalability, efficiency, and ability to support the growing number of Internet-connected devices. Cybersecurity professionals should understand both protocols, as they are commonly encountered in modern enterprise environments.

3.5 Task 05 – OSI Reference Model

Objective

The Open Systems Interconnection (OSI) Reference Model is a conceptual framework that standardizes how data is transmitted between devices over a network. It divides network communication into seven distinct layers, each responsible for a specific set of functions.

OSI Reference Model

Layer	Layer Name	Primary Function	Common Protocol(s)	Typical Device	Real-World Example
7	Application	Provides network services directly to end-user applications.	HTTP, HTTPS, FTP, SMTP	Proxy Server	Opening a website in a web browser
6	Presentation	Translates, encrypts, compresses, and formats data.	SSL/TLS, JPEG, MPEG	Gateway	Encrypting data before online banking
5	Session	Establishes, manages, and terminates communication sessions.	NetBIOS, RPC	Gateway	Maintaining a video conference session
4	Transport	Ensures reliable data delivery, segmentation, and error recovery.	TCP, UDP	Firewall	Downloading a file without corruption
3	Network	Determines the best path for packet	IP, ICMP, IPSec	Router	Sending data across the Internet

Layer	Layer Name	Primary Function	Common Protocol(s)	Typical Device	Real-World Example
		forwarding using logical addressing.			
2	Data Link	Controls node-to-node communication using MAC addresses.	Ethernet, PPP, ARP	Switch	Delivering data to another computer on the same LAN
1	Physical	Transmits raw bits across physical media.	Ethernet Physical Standards, USB	Cable, Hub, Repeater	Data transmitted over an Ethernet cable

Mnemonic for Remembering the OSI Layers

Top to Bottom (Layer 7 → Layer 1)

All People Seem To Need Data Processing

- Application
- Presentation
- Session
- Transport
- Network
- Data Link
- Physical

importance of the OSI Model in Cybersecurity

- The OSI Model enables cybersecurity professionals to:
- Understand how data moves through a network.

- Identify where attacks occur.
- Troubleshoot communication problems efficiently.
- Configure network security devices appropriately.
- Analyze incidents by determining the affected network layer.

Conclusion

- The OSI Reference Model provides a structured framework for understanding network communication. Mastering its seven layers is essential for cybersecurity analysts, network engineers, and system administrators because it supports effective troubleshooting, secure network design, and incident response.

4. Challenges Encountered

During the completion of this assignment, several practical and theoretical challenges were encountered. These challenges provided valuable learning opportunities and contributed to a deeper understanding of computer networking concepts.

Challenge 1: Understanding Network Configuration

Initially, interpreting the output of the **ipconfig /all** command was challenging because it displays numerous networking parameters. Additional study was required to understand the purpose of IPv4 addresses, MAC addresses, subnet masks, DNS servers, and default gateways.

Challenge 2: Analyzing Command Outputs

The outputs generated by networking commands such as **ping**, **tracert**, **nslookup**, and **arp** contained technical information that required careful analysis. Learning how each command contributes to troubleshooting improved my understanding of network diagnostics.

Challenge 3: Differentiating Network Devices

Routers and switches perform related functions within a network, making it initially difficult to distinguish their roles. Through research and practical examples, I learned how routers operate at the Network Layer using IP addresses, while switches operate primarily at the Data Link Layer using MAC addresses.

Challenge 4: Understanding IPv6

IPv6 addressing appeared more complex than IPv4 because of its hexadecimal notation and significantly longer address format. Studying IPv6 addressing rules and examples improved my understanding of modern Internet protocols.

Challenge 5: Memorizing the OSI Model

Remembering the correct order of the seven OSI layers required consistent revision. Using mnemonic techniques and relating each layer to real-world networking devices made the model easier to understand and remember.

Lessons Learned

These challenges strengthened my analytical thinking, troubleshooting abilities, and technical documentation skills. They also highlighted the importance of continuous learning in the field of networking and cybersecurity.

5. Key Learning Outcomes

This assignment significantly improved my understanding of computer networking fundamentals through both theoretical study and practical exercises. The following outcomes were achieved:

- Developed a clear understanding of computer networking concepts and communication processes.
- Learned how to identify and interpret important network configuration parameters using the **ipconfig /all** command.
- Gained practical experience using **ping**, **tracert**, **nslookup**, and **arp** to troubleshoot network connectivity and diagnose communication issues.
- Understood the functional differences between routers and switches and their roles within modern network infrastructures.
- Compared IPv4 and IPv6 addressing schemes, recognizing their characteristics, advantages, limitations, and future applications.
- Mastered the seven layers of the OSI Reference Model and understood the responsibilities, protocols, and networking devices associated with each layer.
- Improved technical reporting, documentation, and professional presentation skills through the preparation of this report.
- Strengthened problem-solving abilities by analyzing networking concepts and interpreting command-line outputs.

6. Conclusion

Computer networking forms the foundation of modern information technology and cybersecurity. Understanding how devices communicate, how data travels across networks, and how network infrastructure operates is essential for protecting digital assets and maintaining reliable communication systems.

Throughout this assignment, I gained practical experience using fundamental networking tools and commands to examine network configurations, verify connectivity, resolve domain names, trace communication paths, and analyze Address Resolution Protocol (ARP) entries. These activities provided valuable insight into how network administrators and cybersecurity professionals diagnose and troubleshoot network-related issues.

The comparison of routers and switches demonstrated the distinct roles each device plays within network infrastructures, while the study of IPv4 and IPv6 highlighted the evolution of Internet addressing and the importance of scalable communication technologies. Additionally, understanding the OSI Reference Model provided a structured framework for analyzing network communication and identifying potential security concerns across different layers.

Overall, this assignment strengthened my technical knowledge, practical networking skills, analytical thinking, and documentation abilities. These competencies provide a solid foundation for future studies in cybersecurity, network administration, digital forensics, penetration testing, and security operations. The knowledge gained during this engagement will continue to support my professional development as I pursue a career in cybersecurity.

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